Yikuan Li

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Education

University of Warwick Sep. 2024 – Jun. 2027 (expected)
BSc Mathematics and Statistics Coventry, UK

- **Year 1:** 88.8/100 (first class; top 10 of 269). **Year 2:** 88.5/100 (first class; top 10 of 257).
- **GPA equivalent:** 4.0/4.0. **Academic Excellence Prize** in Years 1 and 2.
- **Selected coursework:** Optimization (MSc-level), Algorithms, Java Programming, Analysis I–III, Probability & Statistics, Stochastic Processes, Multilinear Algebra, Multivariable Analysis, ODEs & PDEs.

Research Interests

Efficient pre-training and post-training of large language models, with an emphasis on optimization and parameter-efficient fine-tuning (PEFT).

Publications and Preprints

Are Parameter-Efficient Fine-tuning Methods Really Different?
Yikuan Li, Pinyan Lu, and Fanghui Liu. [arXiv][code]

Research Experience

Parameter-efficient fine-tuning · FLINT Lab Jun. 2025 – Present
Research with Prof. Fanghui Liu DCS, Warwick; SJTU

- Compared 10+ orthogonal matrix parameterizations for parameter-efficient fine-tuning in terms of parameter efficiency, expressiveness, and runtime overhead.
- Tested initialization strategies and measured downstream task performance and retention of pretrained capabilities. This work led to the preprint *Are Parameter-Efficient Fine-tuning Methods Really Different?*

Post-training of large language models · RAISE Lab Jun. 2026 – Present
Research with Dr. Paris Giampouras DCS, Warwick

- Explore evolution strategies (ES) for post-training large language models at scale.
- Investigate how to improve ES efficiency with limited GPUs by reducing the number of evaluations that must run sequentially.

Understanding Gradient Orthogonalization in the Muon Optimizer Oct. 2026 – Present
Third-year essay · Supervisor: Dr. Clarice Poon WMI, Warwick

- Use optimization theory to study why gradient orthogonalization may improve training with Muon.

Independent Online Coursework

Completed lectures and assignments independently.
Stanford: CS229 (Machine Learning), CS231N (Computer Vision), CS224N (NLP with Deep Learning).
MIT: 18.S096 (Matrix Calculus for ML and Beyond), 6.S184 (Flow Matching and Diffusion Models).

Technical Skills

Programming: Python, SQL, MATLAB, R, Java
Machine learning: NumPy, PyTorch, Hugging Face Transformers, PEFT
Tools: Git, Linux, Bash, Slurm (GPU workloads), LaTeX
Languages: English (fluent), Mandarin (native), Cantonese (fluent)

Additional Experience

Flywheel May 2024 – Aug. 2024
Data Analyst Intern Hangzhou, China

- Developed time-series forecasting pipelines using Prophet, SARIMA, and LSTM models to predict gross merchandise value (GMV) around major promotional events.
- Automated data consolidation and reporting in Python, reducing manual processing time by approximately 50%.